Measurement of $\nu_\mu + \bar{\nu}_\mu$ Charged-Current Multi-Pion Production on Argon with the MicroBooNE Detector in the NuMI Beam

Technical Note: the Two-Charged-Pion Final State (`CC1mu2piBDT`)

Draft 0.6, 2026-10-17

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Abstract

We describe the extension of the MicroBooNE NuMI $\nu_\mu + \bar{\nu}_\mu$ charged-current single-charged-pion analysis to final states with two charged pions and any number of nucleons, CC2$\pi^{\pm}Xp$. The selection follows the strategy of the BNB CC2$\pi^{\pm}Np$ analysis [3]: a four-class particle classifier (muon, pion, proton, other) on every track, the assignment of the tracks to one muon and N pions by the largest summed log-probability, and an event classifier on the assigned probabilities. On the full NuMI exposure (7.766×10^{20} POT in forward and 1.1082×10^{21} POT in reverse horn current) it selects 13% of the signal at 51% purity in the combined configuration, against 18% at 26% for the selection of the first draft of this note. Without a constraint the uncertainty of the one-bin total is dominated by the flux, amplified by the subtraction of a background as large as the signal, and by the detector variations. Four control samples defined from the classifier outputs condition the covariance of the extraction framework, partly through the signal they contain: in the combined configuration the one-bin total carries 15.2% with the constraint (15.4% with the MCS-calibration term) and 45% without, with data statistics then the largest term; ensembles of pseudo-experiments return the expected truth within 1.5%, with a conservative statistical term, and two reweighted interaction models within 0.2σ. In the opened control regions of the single-pion analysis, calibrating the MCS momentum of exiting tracks as in the single-pion analysis brings the classifier inputs much closer to the beam-on data; with the systematic covariance of the prediction the class probabilities, the MCS momentum and the event score agree with the data, while the LLR score of long protons stays outside ($p = 0.014$) and follows the recombination variation of the detector term. Three or more pions are not measurable at this exposure and serve as a control sample. The signal region is blind, and every result is extracted from pseudo-data.

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1 Introduction

Charged-current (CC) production of multiple charged pions on nuclei sits at the transition between the resonant (RES) and deep-inelastic (DIS) regimes of neutrino–nucleus scattering and is strongly shaped by the intranuclear hadronic cascade. A ν_μ interaction that produces several pions at the primary vertex may have some of them absorbed, charge-exchanged or produced in re-scattering before they leave the nucleus, so the observed charged-pion multiplicity is a direct probe of final-state interactions (FSI). Multi-pion final states are also a background to exclusive single-pion and pion-less measurements, so a measurement of their rate on argon is useful both as a cross-section result and as a constraint on the interaction model used across the MicroBooNE programme.

This note documents the extension of the NuMI $\nu_\mu + \bar{\nu}_\mu$ CC1$\pi^\pm$ analysis [4] to two charged pions. The first draft of this note (2026-08-13) defined the selections `CC1mu2pi` and `CC1mu3pi` as subselections of the single-pion selection `CC1mu1piXp` with a looser pion identification [1, 2], on the Run-1 forward-horn overlay only. Since then the analysis has been rebuilt on the strategy of the BNB CC2$\pi^\pm Np$ analysis [3] and on the current single-pion release: a baseline on the full exposure (Section 4), a particle classifier (Section 5), the track assignment and an event classifier (Section 6), a study of the detector term (Section 7), control samples and a background constraint (Section 8), the extraction of the one-bin total through the framework (Section 9), and the comparison of its inputs with beam-on data in the opened control regions, with the calibration of the MCS momentum and the selection variants (Section 10). The samples, normalisation, systematic universes and Wiener-SVD unfolding [10] are those of the single-pion analysis.

The decisions that fix the scope are (D4 and D5, on NuGraph inputs and a NuMI NuWro sample, are recorded in the plan):

- D1: the signal is inclusive in nucleons, CC2$\pi^\pm Xp$, with a subsample with at least one proton for comparison with the BNB analysis.
- D2: a per-pion momentum threshold of 0.10 GeV/c for the signal, totals and angles; pion-momentum bins only above 0.175 GeV/c, where the range estimator is unbiased.
- D3: the two-pion signal region is blind. Part of it lies in the multi-π control region of the single-pion analysis, which has been compared with beam-on data.
- D6: the truth requirement $\theta(\mu, \text{leading } \pi) < 2.6$ rad has a reconstructed counterpart on the assigned pions.
- D7: no exclusivity cut against the blind single-pion signal region; the overlapping events are opened only after the single-pion signal region.
- D8: the measurements stop at two pions. Events with three or more pion candidates are a control sample of the two-pion background.

2 Signal definition

The signal is a ν_μ or $\bar{\nu}_\mu$ charged-current interaction with its vertex in the fiducial volume of the single-pion analysis, one muon with $p_\mu > 0.15$ GeV/c and exactly two charged pions leaving the nucleus (π^+ or π^-; the pion charge is not measured), counted at any momentum and both with $p_\pi > 0.10$ GeV/c, so that an event with a third charged pion below the threshold is not signal; no neutral pion, kaon or heavier meson, and any number of nucleons. The opening angle between the muon and the leading (hardest) pion is below 2.6 rad, as in the single-pion definition. The subsample with at least one proton above 0.3 GeV/c is 81% of the signal.

Pion-momentum threshold. In a two- or three-pion event the sub-leading pions are softer than the leading one, and the softest pion sets how often the full topology can be reconstructed. On the Run-1 forward-horn overlay the mean leading (softest) true pion momentum is 0.52 (0.27) GeV/c for two pions and 0.57 (0.21) GeV/c for three. The efficiency of the first-draft

selection, evaluated on the signal whose softest pion lies above a threshold $p_\pi^{\min}$, is given in Table 1.

Table 1: Signal efficiency of the first-draft selections as a function of a per-pion momentum threshold $p_\pi^{\min}$ applied to the softest pion, Run-1 forward-horn overlay. "Kept" is the fraction of the $p_\pi > 0$ signal retained; Eff. is the efficiency within the restricted phase space.

	Two pions		Three pions	
$p_\pi^{\min}$ [GeV/c]	Kept	Eff. [%]	Kept	Eff. [%]
0	1.00	16.3	1.00	6.4
0.10	0.90	17.1	0.82	6.2
0.175	0.70	18.9	0.49	8.1
0.25	0.43	20.2	0.26	5.6

Raising the threshold changes the efficiency little: with every pion above 0.175 GeV/c about four in five events are still lost, so the efficiency is limited by the reconstruction of several tracks from one vertex, not by the soft pions. The threshold of 0.10 GeV/c, the pion tracking turn-on, removes the unreconstructable tail (10% of the two-pion signal) and keeps 90% of it; 0.175 GeV/c would keep 65% of the two-pion signal in the combined configuration. Pion-momentum bins start at 0.175 GeV/c (D2).

3 Samples, exposure and normalisation

The samples and normalisation are those of the single-pion release: fourteen ν_μ overlay files (forward horn current Runs 1, 2, 4, 5; reverse horn current Runs 1, 2, 3, 4a, 4b, 4c), seven beam-off and seven dirt samples, each run normalised to its recorded exposure, 7.766×10^{20} POT in forward horn current (FHC), 1.1082×10^{21} POT in reverse horn current (RHC) and 1.8848×10^{21} POT combined. The detector variations are the Run-4 FHC and RHC samples, a central-value sample and eight variations each (light yield down and Rayleigh scattering, recombination, space charge and four wire-response variations). The beam-on data are not used: the pseudo-data are Poisson throws, run by run, of the overlays and the dirt at the data exposure, to which the framework adds the beam-off prediction. Generator predictions with the two-pion signal definition have not yet been produced.

4 Baseline on the full exposure

`CC1mu2pi` and `CC1mu3pi` inherit the `CC1mu1piXp` selection (slice, fiducial volume, topology and muon candidate) with a looser pion identification: vertex distance up to 9.5 cm, the pion BDT of the first draft in place of the single-pion identification, up to two uncontained pion candidates, no shower veto and no single-pion opening-angle cut. Following D6 they now require $\theta(\mu, \text{longest pion candidate}) < 2.6$ rad. The selections store every counted candidate (containment, length, range and MCS momenta, direction, identification scores and the backtracked truth), the true pions, the proton and kaon counts and a truth topology class. On a 30 000-event slice every branch of the single-pion selection is identical before and after these changes. Table 2 gives the baseline at the full exposure, beam-off and dirt included.

Table 2: Baseline of the first-draft selections at the full exposure, with the reconstructed opening-angle requirement. Purities include beam-off and dirt.

	Generated signal	Selected signal	Selected total	Efficiency	Purity
Two pions, FHC	1 210	221	836	18.2%	26.4%
Two pions, RHC	1 672	303	1 168	18.1%	26.0%
Two pions, combined	2 882	524	2 004	18.2%	26.1%
Three pions, FHC	204	21.7	203	10.6%	10.7%
Three pions, RHC	347	32.8	314	9.5%	10.4%
Three pions, combined	551	54.5	517	9.9%	10.5%

These numbers replace the raw Run-1 counts of the first draft, whose "670 (66) events at full exposure" were overlay counts at the simulated exposure. The largest two-pion backgrounds are single-pion events with a second counted candidate (22% of the sample), events with a neutral pion (21%), three or more charged pions (7%), neutral-current events (6%) and beam-off (2.5%). In selected signal events only 60% have both counted candidates backtracked to true pions, and the longest candidate is the leading pion in 52%, so observables ordered by pion cannot be used with this candidate choice. The three-pion sample is 10.5% pure and its largest background, 30%, is a two-pion event with a third counted candidate.

5 Particle classifier

A four-class classifier (muon, pion, proton, other) is trained with XGBoost [5] on the tracks of the pion-candidate pool of the selection (generation 2, track score ≥ 0.3), dumped from all fourteen overlays (3.1 million tracks). Tracks are labelled by their backtracked particle; tracks without a truth match, mostly cosmic rays of the overlay, are the class "other". Training uses run periods 1, 2 and 4 of both horn modes; the held-out periods are RHC Run 3 and FHC Run 5. The beam-on, beam-off and dirt ntuples of the older production (Runs 1–3) lack twenty of the track branches every overlay carries (the Bragg-pion likelihoods, truncated dE/dx, deflection, end spacepoints, hit and descendant counts), so the deployed models use the 43 inputs every sample carries. The model is evaluated in the selection through ROOT's `RBDT` [6], which reproduces XGBoost exactly.

Table 3: ROC area of pion identifications on the two-pion candidate pool, held-out runs, tracks with a truth match (backtracked purity above 0.5).

Pion identification	π vs rest	π vs p	π vs μ
Particle classifier (63 inputs)	0.954	0.985	0.895
Pion BDT of the first draft, as applied	0.888	0.945	0.804
Pion BDT of the single-pion analysis	0.818	0.961	0.558
MIP BDT	0.718	0.957	0.271
LLR score	0.670	0.893	0.251

The gain is the same in FHC and RHC and for π^+ and π^-. With the unmatched tracks as the fourth class, at the pion efficiency of the first-draft BDT (91.8%) the purity of the counted pion candidates rises from 27.6% to 43.3% and the fraction of protons passing falls from 13.4% to 4.1% (63 inputs). With the 43 inputs every sample carries, the pion ROC area on the full pool, unmatched tracks included, is 0.938 (0.946 with all inputs, 0.868 for the first-draft BDT). The pion efficiency does not depend on the true pion momentum more than that of the first-draft BDT and is higher at threshold. Muon against pion is the hardest separation (0.895; about a third of the true muons in the pool still pass).

Used as the pion identification of the single-pion selection at the same pion efficiency, the 43-input classifier raises the purity of the counted tracks from 52.6% to 64.8% and, at event level on the held-out runs, the selection purity from 58.0% to 64.7% for 5% of the signal. This is a study for a later revision of the single-pion analysis.

The LLR-and-geometry classifier. Section 7 shows that the detector variations act on the calorimetric inputs. A second model drops the Bragg likelihoods, χ^2 scores, PIDA and calorimetric energies, and keeps the LLR score (summed and per plane), length, track score, vertex distance, start and end containment, range and MCS momenta, the proton-hypothesis energy, the daughter counts and the per-plane hit flags. Its pion ROC area is 0.892 against 0.926 for the deployed model on the same tracks, and at 80% pion efficiency it passes 9.0% of the protons against 3.9%. It is the particle classifier of `CC1mu2piBDT`.

6 Track assignment and event classifier

Assignment. In events with one neutrino slice and the vertex in the fiducial volume, the tracks of the pion pool with track score ≥ 0.5, the muon candidate included, carry the four class probabilities. For the hypothesis "one muon, N pions, the other tracks protons or other" the log-likelihood is

$$L_N = \log P_\mu(\mu) + \sum_{\text{pions}} \log P_\pi + \sum_{\text{rest}} \log\left(1 - P_\mu - P_\pi\right), \tag{1}$$

where $1 - P_\mu - P_\pi = P_p + P_{\text{other}}$ and every probability is floored at 10^{-6} before the logarithm; every choice of the muon and the N pions among at most eight tracks is enumerated, and the best and the runner-up are kept for $N = 1, 2, 3$. In the selected signal of the held-out periods the assigned muon is a true muon in 89% of events and each assigned pion a true charged pion in 88%; all three are in 76%. In the preselected signal, however, only 30% of the events have the muon and both pions as tracks, so in most signal events no assignment can be fully right.

Event classifier. An XGBoost classifier separates the two-pion signal from everything else on 33 features: the probabilities of the assigned tracks, $L_2 - L_1$, $L_2 - L_3$, the margin over the runner-up, the class sums and the counts above 0.5 over all tracks, the track and shower counts, the topological score and CosmicIP. Its domain is one slice, the vertex in the fiducial volume, the software trigger, at least three tracks and $\theta(\mu, \text{longest assigned pion}) < 2.6$ rad (D6). It is trained on run periods 1, 2 and 4 of both horn modes (206 152 events, 13 558 signal) with the nominal mixture of overlay, dirt and beam-off and the central-value weights, and tested on FHC Run 5 and RHC Run 3 (104 814 events, 7 253 signal), which the particle classifier did not see either. These periods also choose the working point below, so they are validation periods rather than an untouched test sample. The ROC area on them is 0.887; adding the lengths, vertex distances, containment and opening angles of the assigned tracks raises it to 0.898, a gain that disappears in the total uncertainty, so these kinematic features are not used. The largest gains come from the summed pion probability, L_2, the number of tracks with $P_\pi > 0.5$, the topological score and the number with $P_\mu > 0.5$.

Working point. The threshold is chosen on the expected total uncertainty of the one-bin total, with the test-period yields projected to the full exposure: data statistics, flux ($17\% \times (1 + B/S)$), POT and targets, cross section, re-interaction, detector (the eight Run-4 variations, FHC and RHC correlated) and simulation statistics (Table 4). With statistics and flux alone the minimum lies at 3.9% efficiency; the detector term rises steeply above a score of 0.95, and with all terms the minimum is at 0.92, flat between 0.90 and 0.94 within the statistics of the detector-variation samples.

Table 4: Expected uncertainty of the one-bin total (combined configuration) for the first-draft selection, the same with the particle classifier as pion identification (`CC1mu2piNewPID`), and the event classifier with the deployed and with the LLR-and-geometry particle classifier at score > 0.92. The last row is the evaluation of Section 8, with the flux from the PPFX universes [9] instead of $17\% \times (1 + B/S)$. The totals also contain the POT, target and simulation-statistics terms, which are not listed.

Selection	Eff.	Purity	Stat.	Flux	Cross sect.	Reint.	Detector	Total
`CC1mu2pi`	18.2%	26%	8%	65%	47%	17%	34%	89%
`CC1mu2piNewPID`	15.8%	35%	8%	48%	27%	13%	21%	61%
Event classifier, deployed	17.1%	56%	6%	31%	12%	5%	33%	47%
Event classifier, LLR and geometry	13.0%	51%	7%	35%	17%	6%	21%	46%

The cross-section term is the model dependence of the background and falls with the purity; the model dependence of the efficiency stays below 6%. In the combined configuration the signal region of the LLR-and-geometry selection holds 738 predicted events at full exposure, 378 of them signal; the largest backgrounds are single-pion events (15% of the sample) and three or more charged pions (13%). Of the selected events 14% lie in the blind single-pion signal region (D7) and 12% in the opened multi-π control region of the single-pion analysis (D3).

Selection requirements. Table 5 lists the requirements of `CC1mu2piBDT`. The cuts of the single-pion selection after its preselection (topological score above 0.67 and the muon-candidate, pion-containment, wire-gap, shower and track-multiplicity requirements) are not applied; the topological score, CosmicIP and the track and shower counts enter only as features of the event classifier.

Table 5: Requirements of `CC1mu2piBDT`. Cuts 1–9 define the domain of the event classifier; the signal region adds cut 10. The sidebands (Section 8) are the events that pass cuts 1–9 and fail cut 10, assigned in the order listed; in the extraction they exclude the events selected by the single-pion analysis.

	Cut	Requirement
Event		
1	Neutrino slices	exactly one
2	Reconstructed vertex	inside the fiducial volume of the single-pion analysis, outside the dead-wire region in z
3	Software trigger	passed
Track pool: direct neutrino daughters		
4	Track score	≥ 0.5
5	PID inputs	LLR score in $(-1, 2)$; proton, muon and MIP Bragg likelihoods in $(0, 500)$; length > 0
6	Ranking	by $\max(P_\mu, P_\pi)$ of the LLR-and-geometry particle classifier; at most eight tracks kept
7	Pool size	at least three tracks
Assignment and event classifier		
8	Assignment	the muon and two pions that maximise L_2
9	Opening angle	$\theta(\mu, \text{longer assigned pion}) < 2.6$ rad (D6)
10	Event score	> 0.92: signal region
Sidebands: cuts 1–9, score ≤ 0.92		
	Three-pion-like	at least four tracks and $L_3 > L_2$
	Proton-like	an assigned pion with $P_p > 0.5$
	Score band	$0.5 <$ score ≤ 0.92
	Low score	score ≤ 0.5

7 The detector term

At the working point of the deployed classifier the detector term, 33%, is the largest. A paired bootstrap over the physical events, with one Poisson weight per event shared by the central-value sample and the eight variations, attributes 7.7% of it to the statistics of the variation samples. It comes from the recombination (+22%) and wire-response variations (angle yz +18%, x +12%, yz +8%); the light-yield, space-charge and angle-xz variations are consistent with zero. At track level these variations move the χ^2 scores, PIDA, the Bragg likelihoods and the calorimetric energies by up to half of their spread for protons, the LLR score by at most 0.04 and the geometric inputs not at all. At event level they lower the selected signal by about 8% and the background by 10–18%, and the detector term splits into an efficiency part (14.3%), which no background constraint can remove, and a background part (17.8%).

Table 6: Variants of the particle classifier and of the event features (combined configuration), at the best threshold of each. The detector term is split into its efficiency and background parts.

Particle classifier	Event features	Score >	Eff.	Purity	Detector	(eff., bkg.)	Total
Deployed	PID set	0.92	17.1%	56%	32.7%	(14.3, 17.8)	47%
Deployed	full set	0.94	14.4%	62%	33.0%	(18.2, 14.8)	45%
Without χ^2, PIDA, energy	PID set	0.93	14.4%	57%	35.1%	(14.9, 19.0)	48%
LLR and geometry	PID set	0.92	13.0%	51%	21.4%	(7.8, 15.2)	44%
LLR and geometry	full set	0.93	12.0%	55%	20.3%	(8.4, 12.9)	42%

No variant removes the detector term, and the totals lie within the 3% scatter of the scans. The LLR-and-geometry classifier halves the efficiency part (14.3% to 7.8%), the part a constraint cannot reach, and is adopted. Training the event classifier or the particle classifier with the variation samples did not help.

For comparison, the BNB analysis quotes a detector term of 11.0% for the predicted selected event count, from seven Run-4d variations without the wire-angle ones [3]. In the same convention the LLR-and-geometry selection has 10.6% with all eight variations and 8.4% with those seven; the terms of Table 4 are those of the extracted total, in which the background subtraction and the efficiency correction amplify the change of the predicted events by a factor of two to four. The recombination variation leads in both analyses.

8 Control samples and the background constraint

The flux term scales the signal and the background together, and the background is as large as the signal. Control samples that share the background composition constrain it through the conditional covariance of the extraction framework (`ConstrainedCalculator`): the covariance of the predicted counts in the signal region and the sidebands, built from every source, is conditioned on the sidebands, and the background estimate moves with the sideband data. Four sidebands are defined from the classifier outputs, exclusive and in this order inside the two-pion domain (Table 5): three-pion-like (at least four tracks and three pions more likely than two; the control sample of D8), proton-like (an assigned pion with $P_p > 0.5$), event score between 0.5 and the working point, and event score below 0.5. The sidebands leave out the events of the blind single-pion signal region, so they can be opened without it (Table 7).

Table 7: Signal region and sidebands of the LLR-and-geometry selection, combined configuration, predicted events at full exposure.

Region	Events	Signal	Purity	Largest classes
Signal region (score > 0.92)	738	378	51%	CC1π 15%, CC$\geq 3\pi$ 13%
Three-pion-like	282	39	14%	CC$\geq 3\pi$ 26%, CCπ^0 18%, beam-off 15%
Proton-like	15 160	431	3%	CC0π 40%, CCπ^0 16%, CC1π 15%
Score 0.5–0.92	4 397	669	15%	CCπ^0 27%, CC1π 21%
Score < 0.5	10 166	143	1%	beam-off 45%, CC0π 11%, CCπ^0 11%

A first evaluation outside the framework, with the formalism of `ConstrainedCalculator`, gives the uncertainties of Table 8. The flux term falls from 35% to 5.5%, since the flux scales the background and the sidebands alike, and the detector term from 21% to 9%, close to its efficiency part. The three-pion-like sample alone, 282 events, halves the total. The data-statistics term grows with a constraint, because the counts of the sidebands enter the background estimate.

Table 8: Uncertainty of the one-bin total (combined configuration) with each sideband constraint, evaluated outside the framework (flux from the PPFX universes, detector taken fractionally).

Constraint	Total	Flux	Cross sect.	Reint.	Detector	Data stat.	MC stat.
None	45.5%	35.4%	16.7%	5.5%	20.6%	7.2%	3.9%
Three-pion-like	24.2%	13.6%	8.3%	4.2%	11.7%	11.4%	6.8%
Proton-like	31.9%	13.0%	15.2%	5.3%	22.9%	7.3%	4.0%
Score 0.5–0.92	22.5%	6.2%	8.4%	5.2%	17.2%	7.7%	4.2%
Score < 0.5	33.0%	13.6%	10.6%	4.9%	26.2%	7.7%	4.4%
All sidebands	16.8%	5.5%	7.0%	4.8%	8.8%	8.8%	5.0%

Closure under alternative models. Pseudo-data from a reweighted interaction model in every region, extracted with the nominal model, return the alternative truth: with all sidebands 0.952 (-0.29σ) under GENIE [7, 8] multisim universe 545 and 0.994 (-0.03σ) under the $\Delta \to N\pi$ angular variation, against 0.979 and 1.012 without the constraint; over the GENIE multisim universes the bias has an rms of 4.5% with the constraint, within 1σ in every universe, and 17.4% without (98% of the universes within 1σ). These models lie inside the variations that build the covariance, so the test checks the procedure, not the model space.

Signal in the sidebands. The score band holds more predicted signal than the signal region (669 against 378 events), and the variations that move the signal move it in every region together, so the conditioning also carries the nominal signal model from the sidebands into the signal region. Treated as an independent source, uncorrelated with the signal region and the backgrounds, so that the sidebands constrain the background only, the signal variations of the sidebands raise the constrained total from 16.8% to 19.3% (flux 5.5% to 9.6%, cross section 7.0% to 8.6%); leaving the score band out of the constraint gives 19.0%, and both together 21.4%. The score band moves the extraction under universe 545 from 1.012 to 0.952; every variant closes within 0.3σ for both models.

9 Extraction through the framework

`CC1mu2piBDT` implements the selection in the framework: the track pool, the class probabilities of the LLR-and-geometry classifier, the assignment, the 33 features, the event score (`RBDT`) and the region; the selected events are those of the signal region. On a 30 000-event slice the features equal those of the Python study exactly, the scores to 6×10^{-8} and the regions in every event, and

`CC1mu1piXp` is bitwise unchanged. The 28 release samples and the 18 detector-variation samples were processed with it. The bin configuration has the signal region as an ordinary bin and the four sidebands as sideband bins in the signal block, so that their signal is predicted. The keyword `Calculator Constrained` in the extraction configuration selects `ConstrainedCalculator`; the extractor propagates the ordinary block of each covariance and, with the constraint, the part that remains after the conditioning, so that the terms by source add up to the constrained total.

Table 9: One-bin two-pion total through the framework, on pseudo-data, without and with the sideband constraint; detector variations taken fractionally and weighted to the current flux (Section 9.1). The last column is the extracted value over the truth of the pseudo-data. The total also contains the POT and target and the simulation-statistics terms, which are not listed.

	Total	Detector	Flux	Cross sect.	Reint.	Data stat.	Unfolded/truth
FHC, no constraint	54.0%	34.2%	37.0%	14.7%	5.2%	10.5%	1.056
FHC, constrained	22.5%	11.4%	6.1%	8.7%	5.9%	14.0%	1.020
RHC, no constraint	41.1%	18.1%	31.5%	15.5%	5.4%	8.7%	1.035
RHC, constrained	16.3%	7.3%	4.9%	5.9%	4.1%	11.1%	1.020
Combined, no constraint	46.0%	25.9%	33.7%	15.0%	5.2%	6.7%	1.044
Combined, constrained	15.7%	8.2%	4.8%	6.6%	5.0%	8.8%	1.021

The constrained combined total, 15.7%, agrees with the evaluation outside the framework (16.8%). With the constraint the data statistics become the largest term, and in the three configurations the pseudo-data realisation returns 1.020–1.021 times its truth.

9.1 Detector-variation samples

The first framework extraction gave a larger detector term than the study outside it, most in RHC (28% against 17%). Per POT, the Run-4 detector-variation central-value samples predict 1.27 (FHC) and 1.51 (RHC) times the event rate of the nominal Run-4 overlays in the fiducial volume, the ratio rising from 1.1 below 0.5 GeV to 1.8–3 above 5 GeV. The samples were simulated with the earlier NuMI flux, the nominal overlays with the current one (Geant v4.10.4): weighted event by event with the ratio of the two fluxes in true energy and angle to the beamline, they predict 0.94 times the nominal rate, flat in energy. The framework built the detector covariance from the absolute difference between each variation and the variation central value, and so carried that ratio. The covariance type `DVfrac` applies the fractional change of each variation relative to the variation central value to the nominal prediction; it is used here and, since version 1.8, in the single-pion release.

Because the earlier flux overweights high energies, where the two-pion signal sits, the variation samples are in addition weighted to the current flux: the selection framework applies an optional per-event weight branch, carried only by reweighted copies of the 18 variation samples. On the fractional detector change of the selected two-pion count the weight raises the FHC term from 17.6% to 20.6% and leaves RHC unchanged (9.3% to 9.1%); for the single-pion selection it changes the term by at most 0.3% absolute. Table 10 shows the three treatments.

Table 10: Detector term and total of the one-bin two-pion total for the three treatments of the detector variations.

	Absolute		Fractional		Fractional, flux-weighted	
	Detector	Total	Detector	Total	Detector	Total
FHC, no constraint	40.8%	58.4%	29.8%	51.3%	34.2%	54.0%
FHC, constrained	15.4%	24.9%	11.9%	22.4%	11.4%	22.5%
RHC, no constraint	28.4%	46.6%	17.9%	41.0%	18.1%	41.1%
RHC, constrained	11.1%	18.5%	7.6%	16.3%	7.3%	16.3%
Combined, no constraint	33.4%	50.7%	23.0%	44.5%	25.9%	46.0%
Combined, constrained	11.9%	17.9%	8.5%	15.7%	8.2%	15.7%

The sideband constraint removes the difference between the fractional treatments: the constrained totals are the same with and without the flux weight. The ratio of a variation to the variation CV, which the fractional treatment transfers to the nominal prediction, changes by at most 0.027 (median 0.003) with the flux weight, for signal and background in the signal region and the score band; above 1 GeV it follows the integral within the statistics of the variation samples (5–6%), the largest trend being the recombination variation of the FHC background (0.76 at 1–2 GeV, 1.01 above 5 GeV).

10 Beam-on data in the control regions

The inputs and the output of the two-pion selection were compared with beam-on data in the four control regions of the single-pion analysis, which are open. The skim keeps only events in those regions and outside the signal regions of both selections, so the two-pion signal region is absent from data and prediction alike. The prediction is the sum of the overlays, beam-off data and dirt; the shape tests remove the normalisation and are made per run period and summed. In the two-pion domain the data exceed the prediction by 10–24% (three-pion-like 1.18, proton-like 1.10, score 0.5–0.92 1.20, score below 0.5 1.24), as outside it (1.12); the largest excess is in the π^0 control region.

MCS momentum. The MCS momentum of the pool tracks is the most mis-modelled input: data over prediction rises to 1.14 at 0.7–0.9 GeV/c and falls to 0.39 at 1.8–1.9 GeV/c (shape $\chi^2 = 2234/19$). This is the effect found for exiting muons in the single-pion analysis [4], an angular noise of reconstructed tracks on the scale of the MCS segments that the simulation lacks, calibrated there from version 1.9 by a quantile map of the MCS momentum in cells of the drift-direction component and the length of the track. The same map is applied here to the classifier inputs of every simulated track whose end leaves the containment volume (`CC1MU1PIXP_MP_MCS_MAP`; without it every output is unchanged, bit for bit). With it the shape χ^2 of the MCS momentum falls to 71.5/19 (data over prediction 0.87–1.04), that of $P(\mu)$ from 1100 to 293, of $P(\pi)$ from 256 to 56 (19 degrees of freedom) and of the summed $P(\pi)$ of the event from 92 to 52 (15), in every period.

LLR of long protons. Among pool tracks longer than 40 cm, those with an LLR score below -0.9 (an LLR sum below -630) are 11 times more frequent in the data than predicted (592 against 52); between -0.9 and -0.8 the data hold 1.33 times the prediction and between -0.8 and -0.5 0.84. They are long contained protons: both ends contained (97%), starting at the vertex, 40 to 100 cm long, true protons in 72% of the predicted ones, in every period and both productions and on all three planes. With the same number of hits they are more proton-like per hit in the data: for contained tracks longer than 40 cm with an LLR score below -0.5 the median LLR sum is -436 to -486 in the Run-4 data and -339 in the simulation. Of the detector

variations the recombination variation reproduces it (median −445; share beyond −0.9 of 0.15, against 0.16–0.23 in the data and 0.015 in the simulation), and the wire-response variations move it the other way. The effect is therefore inside the detector term, of which the recombination variation is the largest part (Section 7).

Event score. With the calibrated simulation the shape χ^2 of the event score is 140/22 (160/22 before). The bin below the cut holds 0.83 of the predicted share, and the recombination variation gives 0.86 in both modes. The lowest bin (0.78) holds control-region CC0π events (97.5%); against the normalisation of that region alone it is at 0.85, the rest coming from the larger excess of the π^0 region in the common normalisation, and the recombination variation gives 0.93–0.94.

With the systematic covariance. The shape tests above carry the data and simulation statistics only. With the covariance of the prediction (flux, cross-section and reinteraction universes, the eight detector variations taken fractionally per horn mode with the statistics of the variation samples, simulation statistics; normalisation removed) the event score gives $\chi^2 = 6.5/11$, the summed $P(\pi)$ 2.0/7, the track $P(\mu)$ and $P(\pi)$ 4.2/9 each and the MCS momentum 6.2/8, against 98.5, 23.6, 48.1, 40.4 and 65.2 with statistics alone in the same, coarser bins. The LLR score stays outside, 20.8/9 ($p = 0.014$); the detector term alone brings it from 398 to 36.

Extraction with the calibrated simulation. The overlays, the dirt and the detector-variation samples were processed again with the calibrated classifier inputs, and the pseudo-data thrown with the same seeds, so that they hold the same events. Table 11 gives the totals. The calibration changes the expected content of the signal region by less than 1% (signal −0.9% in FHC and −0.5% in RHC, all events −0.7% and 0.0%) and the constrained combined total from 15.7% to 15.2%. The classification moves about 5% of the signal-region events in or out, and their pseudo-data counts are Poisson draws made without regard to it; this moves the closure to 1.05–1.09, by +0.9 (FHC) and +2.3 (RHC) times the Poisson spread of the migrating events. The calibration uncertainty is evaluated as in the single-pion analysis: pseudo-data with the classifier inputs calibrated by each of the seven alternative maps, extracted against the response calibrated with the nominal map, in four fully correlated terms: 2.2% of the constrained combined total, 2.3–5.5% in the other extractions and at most 0.24 of their total uncertainty. Each alternative map moves only 4 to 19 of the pseudo-data events in or out of the signal region, and every shift lies within 1.7 times the Poisson noise of those events, so the term is an upper estimate. The calibrated selection is used from here on.

Table 11: One-bin two-pion total with the MCS-calibrated classifier inputs, as Table 9. The calibration term is not part of the total; added in quadrature it gives 15.4% for the constrained combined total.

	Total	Detector	Flux	Cross sect.	Reint.	Data stat.	Calibration	Unfolded/truth
FHC, no constraint	51.3%	30.9%	36.2%	14.6%	5.1%	10.4%	5.5%	1.082
FHC, constrained	21.3%	10.6%	6.2%	7.8%	5.4%	13.5%	5.1%	1.050
RHC, no constraint	40.3%	19.3%	30.1%	15.0%	5.1%	8.5%	2.5%	1.090
RHC, constrained	15.4%	6.6%	4.2%	5.6%	4.2%	10.7%	2.3%	1.070
Combined, no constraint	44.9%	25.6%	32.5%	14.7%	5.0%	6.6%	3.1%	1.087
Combined, constrained	15.2%	8.0%	4.6%	6.3%	4.7%	8.7%	2.2%	1.056

Ensembles. One hundred pseudo-experiments thrown independently from the nominal calibrated simulation (the eight per-run pseudo-data files each) and extracted through the framework return the expected truth within +0.9 to +1.5% (±0.6 to 1.0%) in FHC, RHC and combined, with and without the constraint. Their spread is 0.85 times the quoted data-statistics term without the constraint and 0.74–0.76 with it, so the statistical intervals cover more than nominal

(75–83% for 68%, 96–99% for 95%). Part of the difference with the constraint is the beam-off, which the framework adds to the pseudo-data without fluctuation while the quoted term counts it (45% of the sideband below a score of 0.5); without the constraint, where the beam-off is 0.4% of the signal region, the difference is not understood yet. Fifty pseudo-experiments each from GENIE multisim universe 545 and from the $\Delta \to N\pi$ angular variation, extracted with the nominal response, return their own truth within 0.07σ and 0.17σ of the total uncertainty (the largest: FHC with the constraint, 0.962 ± 0.014 under the $\Delta \to N\pi$ variation).

Selection variants. The working point was scanned again on the constrained combined total (score above 0.85 to 0.96): 0.92 remains the minimum (16.8%), and tighter thresholds reach 58–70% purity at a total of 22–32%. Per horn mode the minimum lies at 0.90 in FHC (25.9%; 27.3% at 0.92, within 2.7 points from 0.85 to 0.95) and at 0.92 in RHC (19.3%). Event-classifier features from the reconstructed showers (energy, vertex distance, track score and direction of the leading shower; with its dE/dx, the second shower and the two-shower mass in a larger set) raise the area under the ROC curve on the test periods from 0.874 to 0.884–0.887 and the purity at a fixed threshold by about two points. In the π^0 control region, against its own normalisation, the energy, vertex distance, track score and direction of the leading shower follow the beam-on data; its dE/dx and the multi-shower features do not (χ^2 of 22–63 for 3 to 6 degrees of freedom). With the calibrated inputs the best constrained totals of the variants, with and without shower features and with the calorimetric particle classifier, are 16.3–17.3% (21.1% for the calorimetric classifier without shower features), against 16.8% for the adopted selection; the detector term of one variant moves by 3–6 points between neighbouring thresholds, which is more than these differences. The selection is kept.

11 Three or more pions

In the two-pion domain, events with at least four tracks in which three pions are more likely than two number 234 at full exposure (combined, with the deployed particle classifier of the Phase-2 study; the three-pion-like sideband of the adopted selection holds 282, Table 7), of which 88 have three or more charged pions, 41 a neutral pion and 35 are two-pion signal. A three-pion event classifier built the same way reaches a ROC area of 0.855, but with all terms every three-pion selection carries a total uncertainty above 100% (`CC1mu3pi` 245%, with the particle classifier 117%, the event classifier at best 105%). Four and five pions are out of reach: combined, 3.1 selected four-pion signal events at 4.8% purity and 0.1 five-pion events, and the probability that a true pion multiplicity is reconstructed in its own bin is 0.84, 0.48 and 0.33 for one, two and three or more pions. The three-pion-like events are therefore the control sample of Section 8 (D8).

12 Status and next steps

- The two-pion one-bin total is extracted through the framework in FHC, RHC and combined, with and without the sideband constraint, with the MCS-calibrated classifier inputs (Table 11).
- The inputs of the LLR-and-geometry classifier and the event score are compared with beam-on data in the opened control regions (Section 10): the MCS momentum is calibrated, and the LLR of long protons follows the recombination variation of the detector term.
- The selection was compared with event-classifier features from the reconstructed showers and with the calorimetric particle classifier on the constrained total (Section 10); no variant improves it beyond the statistics of the detector-variation samples, and the selection is kept.

- Validation through the framework: the ensembles of pseudo-experiments and the alternative-model ensembles are done (Section 10); open are the origin of the conservative statistical term without the constraint and an independent-generator closure, if a NuMI NuWro overlay can be validated.
- Generator predictions with the two-pion signal definition.
- Response and binning of the differential candidates (p_μ, $\cos\theta_\mu$, pion angles with a convention symmetric in the pions, the opening angle between the pions, the proton multiplicity) under the 0.50 migration criterion; the invariant masses remain study products unless they pass it.
- The same sideband constraint for the single-pion analysis, on pseudo-data.

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